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Comparative Analysis of Multiclass Heart Disease Prediction Classification Models using Preprocessing and Feature Selection

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***Abstract*— Prediction of heart disease and diagnosing it is a challenging task in the medical industry. All over the world, heart disease is considered as the deadliest disease in the life of the human being. Most deaths occur if the cause of the disease is not known in advance. The correct prediction of heart disease can save the life of the person. By selecting significant features and applying classification algorithms, heart disease could be predicted in an efficient way. In this paper, the preprocessing techniques like handling missing values, eliminating duplicate rows, and removing outliers have been applied to the UCI repository heart disease dataset. Machine learning classification algorithms such as SVM, Gradient Booster, Random forest, and Decision Tree were implemented in the dataset and the prediction accuracy was compared. The result shows that these models have produced more accuracy after preprocessing and feature selection techniques.**

Keywords*— *Heart disease, multiclass , preprocessing, feature selection, classification algorithms

I. INTRODUCTION

In personalised medicine, the analysis of healthcare data is becoming increasingly essential. Early detection of the disease is important. For example, in any disease, personalised treatment aims to give the proper treatment to the right patient by considering a variety of factors such as age, behaviour, and lifestyle. Modern technologies such as genetics, imaging and lifetime monitoring devices have generated enormous and complex amounts of health data in recent years, allowing researchers to access this data efficiently and make decisions that provide the best treatment for patients.

Heart disease is widely treated as the most deadly disease in human history. During this type of disease, the heart cannot supply enough blood to the remaining organs of the human body to perform their normal functions [1]. On the basis of medical history, professional symptom analysis reports, and physical laboratory findings, invasive techniques are used to identify heart disease problems. However, at the moment of assessment, it is time-consuming, computationally complex and expensive [2]. In the medical industry predicting and diagnosing heart disease is the most difficult task and it is dependent on elements such as symptoms, physical examination and signs of the patient [3-5].

A wide range of Artificial Intelligence (AI), Deep Learning (DL) and Machine Learning (ML) approaches have been employed to efficiently evaluate the vast data in healthcare throughout the last decade. For example, in heart disease diagnosis, a logistic regression-based prediction model was used for automated early detection of heart disease. Medical imaging has also used machine learning techniques to allow automatic detection of object features. A variety of Deep learning algorithms have also been successfully used to forecast the disease at an early stage [6].

Predictive analytics is a type of technology that forecasts future unknowns based on historical data. For predicting the outcome of the existing data, it uses AI, ML and DL techniques. The main purpose of using these methods in health care is to enhance disease detection and classification, generate better predictions and helps personalised medicine, and the same is true for heart failure treatment [7, 8].

In this paper, the different Machine Learning classification methods were applied to the dataset. The performance is evaluated and the accuracy is compared, before and after, applying preprocessing and feature selection steps.

II. HEART DISEASE DATASET

The dataset consists of 920 patient details with 76 attributes. These are collected from Hungary, Cleveland, Switzerland and Long Beach datasets existing in the UCI repository. However, along with the target attribute, only a subset of 14 attributes, namely age and sex of the person, resting blood pressure, type of chest pain, level of cholesterol, and blood sugar in fasting stage, etc. are used. Table 1 lists all 14 attributes with their description.

The embedded sensors in an ECG (Electrocardiogram), by placing it on a patient's chest, help in detecting irregular heartbeat and stroke [9]. Angina is a discomfort caused by a lack of oxygen in the blood supply to the heart muscle. Atypical, typical, asymptomatic, and non-angina pain are the different forms of angina. Heart disease is more likely in people who have high blood pressure and fast heart rate. The risk of coronary heart disease is directly proportional to heart rate. Symptoms of heart disease are usually chest tightness and it extends from the abdomen to the shoulders.

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TABLE 1. HEART DISEASE DATASET FEATURES

FEATURE	DESCRIPTION
age	age in years
sex	sex (1 = male; 0 = female)
cp	chest pain type (Value 1: typical angina, Value 2: atypical angina,Value 3: non-anginal pain and Value 4: asymptomatic)
trestbps	resting blood pressure (in mm Hg on admission to the hospital)
chol	serum cholesterol (less than 200 mg/dL)
fbs	fasting blood sugar > 120 mg/dl (1 = true; 0 = false)
restecg	resting electrocardiographic results (Value 0: normal, Value 1: having ST-T wave abnormality , Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria)
thalach	maximum heart rate achieved
exang	exercise induced angina (1 = yes; 0 = no)
oldpeak	ST depression induced by exercise in comparison with rest state
slope	the slope of the peak exercise ST segment (1: upsloping, 2: flat and 3: downsloping
ca	number of major vessels (0-3) colored by fluoroscopy
thal	Period of exercise test in minutes. It demonstrates the heart status with one of three values (3 = normal; 6 = fixed defect; 7 = reversible defect)
num	diagnosis of heart disease (angiographic disease status) 0 = no risk of heart disease, 1 to 4 = risk of heart disease

III. RESEARCH METHODOLOGIES

The framework of the proposed model is shown in Fig. 1. The model consists of various stages. The first stage of the heart disease prediction model is collecting a dataset from the UCI repository. The existence of duplicate data will reduce the quality of output. Therefore it should be removed before applying the Machine Learning algorithms. In the preprocessing stage, it deals with handling missing values and data reduction. In the third stage, the most significant features are identified using Correlation Coefficient methods. In the next stage, the proposed model trained the dataset using SVM, Decision Tree, Gradient Booster, and Random Forest classifiers. Finally, put the suggested model to the test, evaluating it for accuracy and performance using a variety of performance indicators. An effective heart disease prediction system is depicted in this model.

A. Colab – Python

"Colaboratory", or "Colab", is a product from Google Research. Colab is a Cloud-based Python programming environment that operates in the browser. It's particularly well adapted to data analysis and machine learning models and it also uses graphs and charts to show the results. Python has a library called numpy, which gives lots of functions that are useful when doing Machine Learning models. In addition, Python has scikit, an open-source library, which provides efficient tools for statistical and machine learning modeling.

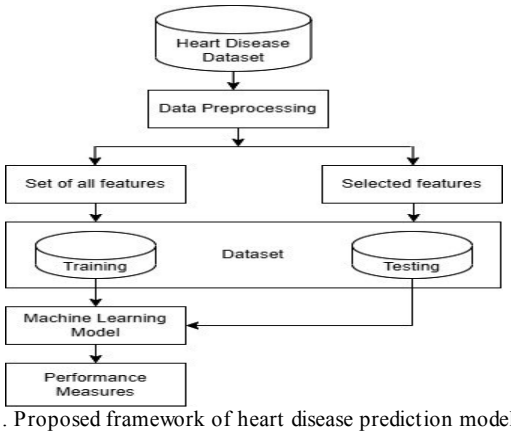


Fig.1. Proposed framework of heart disease prediction model

B. Preprocessing

Preprocessing techniques are useful for processing the raw data and make it suitable for a machine learning model [10]. Data preprocessing methods have been devised and employed, to improve the efficiency of classifier models for heart disease prediction. Data Cleaning is one of the preprocessing techniques, in which it deals with missing data, finding and removing outliers, identifying and eliminating noise, and resolving data inconsistencies.

1) Handling missing values: Because missing data can lead to errors in statistical calculation and misleading visualization, checking the integrity and consistency of data is necessary before processing it. To check whether the dataset has null values, different methods are available. The df.isna( ).sum( ) is used to see how many null values are there for the attributes in a dataset. The Missingno library can also be used to visualize missing data. The horizontal lines shown in Fig. 2 represent the values that are missing. This library is useful for viewing missing values and determining correlations between the missing values in various columns.

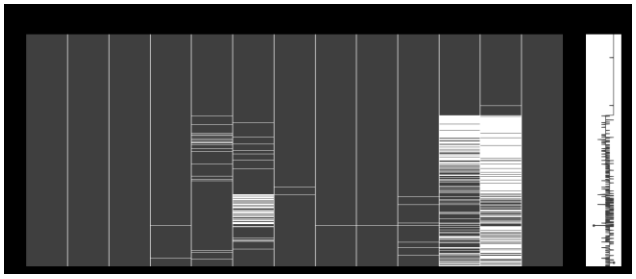


Fig. 2. Columns with missing data using Missingno library

To handle missing data, any one of the following techniques can be used: (i) missing data elimination (2) toleration of missing data and (3) missing data imputation. In the elimination technique, the records that have more missing value attributes are deleted, but in the toleration technique, it is possible to manage data with all missing values. Imputation is the process of substituting missing data with a statistical estimation of those missing data in a data set without loss of any information. There are two categories of imputation

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techniques: statistical imputation and machine learning imputation.

When using Pandas, NaN is automatically substituted for all null values in the selected dataset. The rows and columns with more null values are removed. Then median imputation, which belongs to the category of statistical imputation, is applied to the data set with 867 samples and 13 attributes. Each NaN value is replaced with the column's median value. The data set without null values are shown in Fig.3. The uniqueness of all features is verified. Also the uniqueness of all the rows was verified and duplicate rows are removed from the dataset.

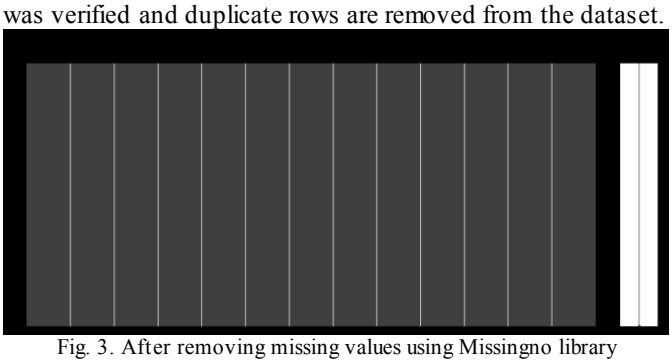


Fig. 3. After removing missing values using Missingno library

2) Identifying and removing Outliers: An outlier is a data point that stands out from the rest of the data. There are two major forms of outlier analysis- univariate (single variable) and multivariate (two or more variables). Outliers can be found using either graphical tools or mathematical functions. We have a boxplot method or a scatter plot method in visualisation tools. A scatter plot is a graph that represents values for two variables in a collection of data using Cartesian Coordinates. An outlier is depicted as a point in a box plot, but other populations are aggregated and displayed as boxes. Fig. 4 shows the outliers in the continuous features such as age, trestbps, chol, thalach and oldpeak.

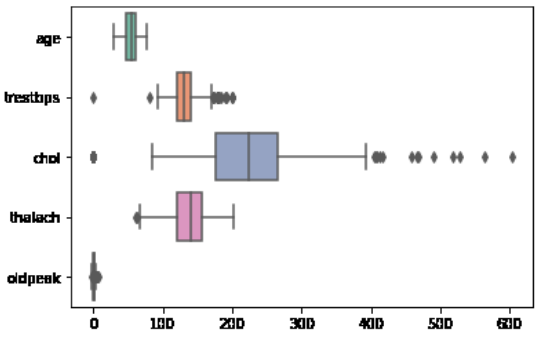


Fig.4. Outliers in continuous features

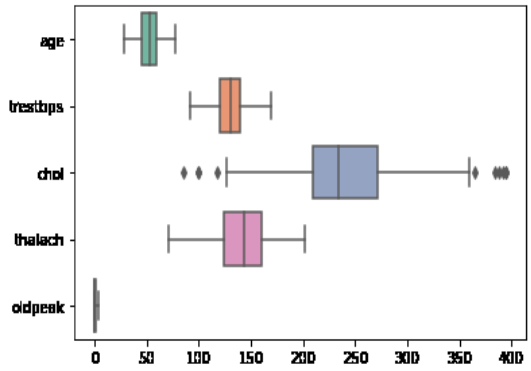


Fig.5. Continuous features after removing outliers

A Z-score or an InterQuartile Range (IQR) score can be used in mathematical functions. The Z-score can be used to find data distributions with a mean of 0 and standard deviation of 1. The difference between upper and lower quartiles is measured by IQR, a statistical dispersion measure. When applying the IQR method, due to data inconsistency, certain samples are deleted from the data set to eliminate outliers. Fig.5 shows continuous features after removing outliers

IV. DISTRIBUTIONS OF ATTRIBUTES

Exploratory Data Analysis is useful to understand the data and their relationship. The bar diagram in Fig.6 shows the disease status in both males and females. From this, we understand males have heart attack more than females.

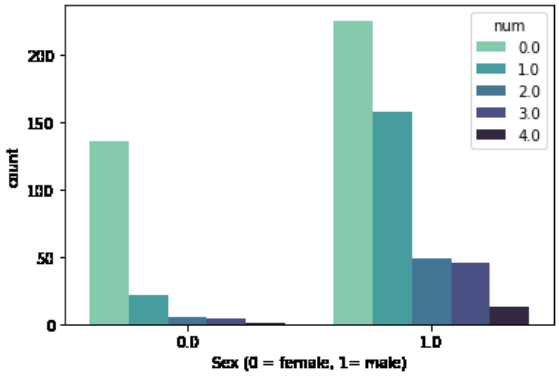


Fig.6. Number of male and female with or without disease

In age distribution, Fig.7 shows that most of the patients are aged between 50 and 60. The diabetes indicator fbs is more than 120 mg/d, the patient is diabetic. In Fig.8, the number for the true class is less than the number for the false class indicating that more patients with heart disease do not have diabetes. This suggests that fbs may not be a key distinguishing factor between patients with or without heart disease.

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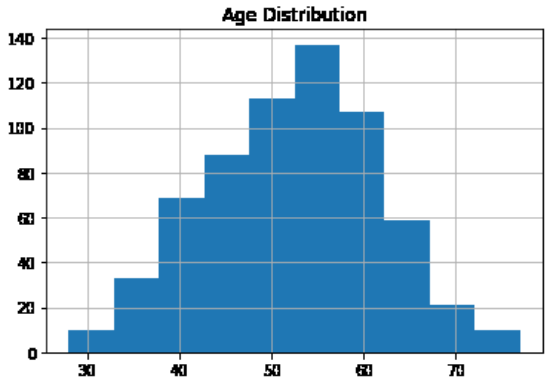


Fig.7. Distribution of age

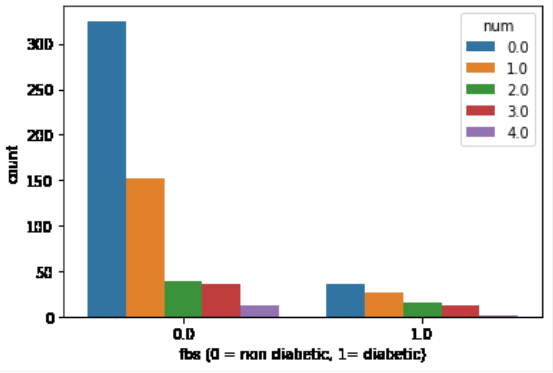


Fig.8. Fasting blood sugar count related to target variable

V. Feature Selection

The purpose of feature selection is always to find the most important heart disease features. Moreover, feature selection helps in the construction of a more accurate model by removing less relevant information, reducing training time and improving learning performance [11]. Pearson's Correlation coefficient is one of the popular feature selection techniques which can be used to measure the association between continuous variables. The Correlation method is applied to 657 samples and 13 distinct features. The more correlated 10 features such as age, sex, cp, restecg, thalach, exang, oldpeak, slope, thal, and num were selected to predict heart disease.

VI. Results and Comparison

The Machine Learning Supervised Algorithms such as Random Forest, Support Vector Machine (SVM), Decision Tree and Gradient Booster applied to the dataset. A decision tree is a supervised learning technique commonly used to solve classification problems. It can be used with categorical or continuous dependent variables. SVM is a supervised learning approach, its main goal is to minimize generalization errors while performing regression and classification tasks. SVM has shown great performance in the medical field in recent years [12, 13] for disease prediction. Gradient boosting classifiers is a collection of machine learning algorithms that build a strong predictor by combining many weak predictors. Random Forest is a machine learning approach that combines numerous classifiers to tackle a complex problem and improve the performance of the model.

TABLE 2. ACCURACY OF MACHINE LEARNING MODELS

Machine Learning Model	Accuracy Before Preprocessing	Accuracy After Preprocessing
Random Forest	57%	67%
SVM	58%	68%
Decision Tree	54%	59%
Gradient Booster	56%	63%

Python was utilized to develop the analytical model using Colab notebooks in this comparative analysis approach. This improves dataset exploration and provides better pattern recognition. Four classifiers are used to assess the performance: random forest, SVM, decision tree, and gradient booster techniques. Table 2 illustrates the accuracy of the model before and after applying preprocessing and feature selection techniques. Fig.9 shows that SVM has the highest accuracy compared to other classifiers.

VII. Conclusion

In the prediction of cardiac disease, machine learning algorithms play a vital role. In this paper, preprocessing and feature selection techniques are applied to ensure data consistency. Different Machine learning algorithms were tried on the selected dataset. The accuracy of the models before and after preprocessing was compared. The result shows that the prediction models have produced more accuracy after implementing preprocessing and feature selection techniques. In the future, feature extraction and boosting techniques will also be tried to improve accuracy.

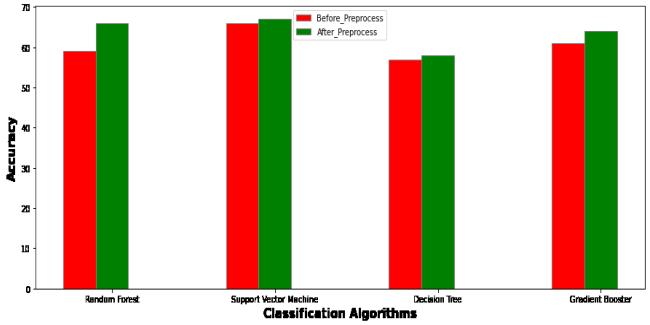


Fig.9. Classification accuracy before and after preprocessing and feature selection

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